

Jaehyung Jung, Ph.D.

Research Scientist

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EXPERIENCES

- **Institute of Robotics and Mechatronics, DLR** July 2026 – Present
Research Scientist
- **Mobile Robotics Lab, Technical University of Munich** Sept. 2023 – June 2026
Postdoctoral Researcher (Supervisor: Prof. Stefan Leutenegger)

EDUCATION

- **Ph.D. Aerospace Engineering, Seoul National University (SNU)** 2023
Thesis: Visual-Inertial Navigation System on Matrix Lie Group with Semantic Objects
- **M.S. Aerospace Engineering, Seoul National University** 2019
Thesis: Self-Calibrated Visual-Inertial Odometry for Rover Localization
- **B.S. Aerospace Engineering, Pusan National University** 2017
Magna Cum Laude (94%)

TECHNICAL SKILLS AND INTERESTS

Expertise: State estimation, Robot mapping

Programming: C++, Python, ROS/ROS2, PyTorch, git

Languages: English (C1, TOEFL: 101/120), German (A1, GOETHE 86/100), Korean (native)

OPENSOURCES

- **OKVIS2-X: Multi-sensor SLAM with volumetric dense mapping** in C++, Python, ROS2, PyTorch 2025
- **Gaussian Mixture Midway-Merge for Object SLAM** in MATLAB 2022
- **Ensemble Visual-Inertial-Odometry** in C++, ROS 2022

SELECTED PUBLICATIONS

- Co-authored 11 publications (290+ citations) in robotics conferences/journals (ICRA, IROS, RA-L, T-RO)
- Please see [GoogleScholar](#) for the full list.

1. Liang Hong, Jiaxin Wei, Simon Schaefer, Stefan Leutenegger, **Jaehyung Jung**, “SpotlessGS: Relightable 3D Gaussian Splatting under Dynamic Illumination for Robotic Perception,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
2. Sebastián Barbas Laina, Simon Boche, Sotiris Papatheodorou, Simon Schaefer, **Jaehyung Jung**, Stefan Leutenegger, “FindAnything: Open-Vocabulary and Object-Centric Mapping for Robot Exploration in Any Environment,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [YouTube](#)
3. Simon Boche*, **Jaehyung Jung***, Sebastián Barbas Laina*, and Stefan Leutenegger (*equal contribution), “OKVIS2-X: Open Keyframe-based Visual-Inertial SLAM Configurable with Dense Depth or LiDAR, and GNSS,” *IEEE Transactions on Robotics*, vol. 41, pp. 6064-6083, 2025. [YouTube](#)
4. **Jaehyung Jung**, Simon Boche, Sebastián Barbas Laina, and Stefan Leutenegger, “Uncertainty-Aware Visual-Inertial SLAM with Volumetric Occupancy Mapping,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [YouTube](#)
5. **Jaehyung Jung** and Chan Gook Park, “Gaussian Mixture Midway-Merge for Object SLAM with Pose Ambiguity,” *IEEE Robotics and Automation Letters*, vol. 8, no. 1, pp. 400-407, 2023. [YouTube](#)
6. **Jaehyung Jung**, Yeongkwon Choe, and Chan Gook Park, “Photometric Visual-Inertial Navigation with Uncertainty-Aware Ensembles,” *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2039-2052, 2022. [YouTube](#)
7. **Jaehyung Jung**, Jaehyuck Cha, Jae Young Chung, Tae Ihn Kim, Myung Hwan Seo, Sang Yeon Park, Jong Yun Yeo, and Chan Gook Park, “Monocular Visual-Inertial-Wheel Odometry using Low-Grade IMU in Urban Areas,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 925-938, 2022.

8. **Jaehyung Jung** and Chan Gook Park, “Object-based Visual-Inertial Navigation System on Matrix Lie Group,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2022. [YouTube](#)
9. **Jaehyung Jung** and Chan Gook Park, “Constrained Filtering-based Fusion of Images, Events, and Inertial Measurements for Pose Estimation,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2020. [YouTube](#)

ACADEMIC ACTIVITIES

- **Co-organizing Workshop in ICRA 2025** 2025
Novel Approaches for Precision Agriculture and Forestry with Autonomous Robots
- **Organizing Workshop in RSS 2024** 2024
Navigation & Mobile Manipulation in Challenging and Cluttered Natural Environments
- **A reviewer in ICRA, IROS, RSS, RA-L, T-RO** 2021 – Present

AWARDS

- **IEEE ICRA Best Paper Award** on Robot Perception 2026
- **Best Paper Award** in Avionics Systems Symposium Korea 2019
- **Best Paper Award** in 33rd Institute of Control, Robotics, and Systems, Annual Conference 2018
- **The 2nd Prize** in CANSAT competition Korea 2016
 - Organized by Ministry of Science and ICT
 - Team “To the Space!”: 2-DOF camera gimbal stabilizer for CANSAT

TEACHING EXPERIENCES

- **TA in Mobile Robotics (TUM)** Theories in perception, control, and path planning 2024, 2023
- **TA in Mobile Robotics Practicals (TUM)** Autonomous flight (Point A-to-B with obstacles) 2024, 2023